



pumas^{AI}



DeepPumas Random effects...

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What are Mixed Effects?

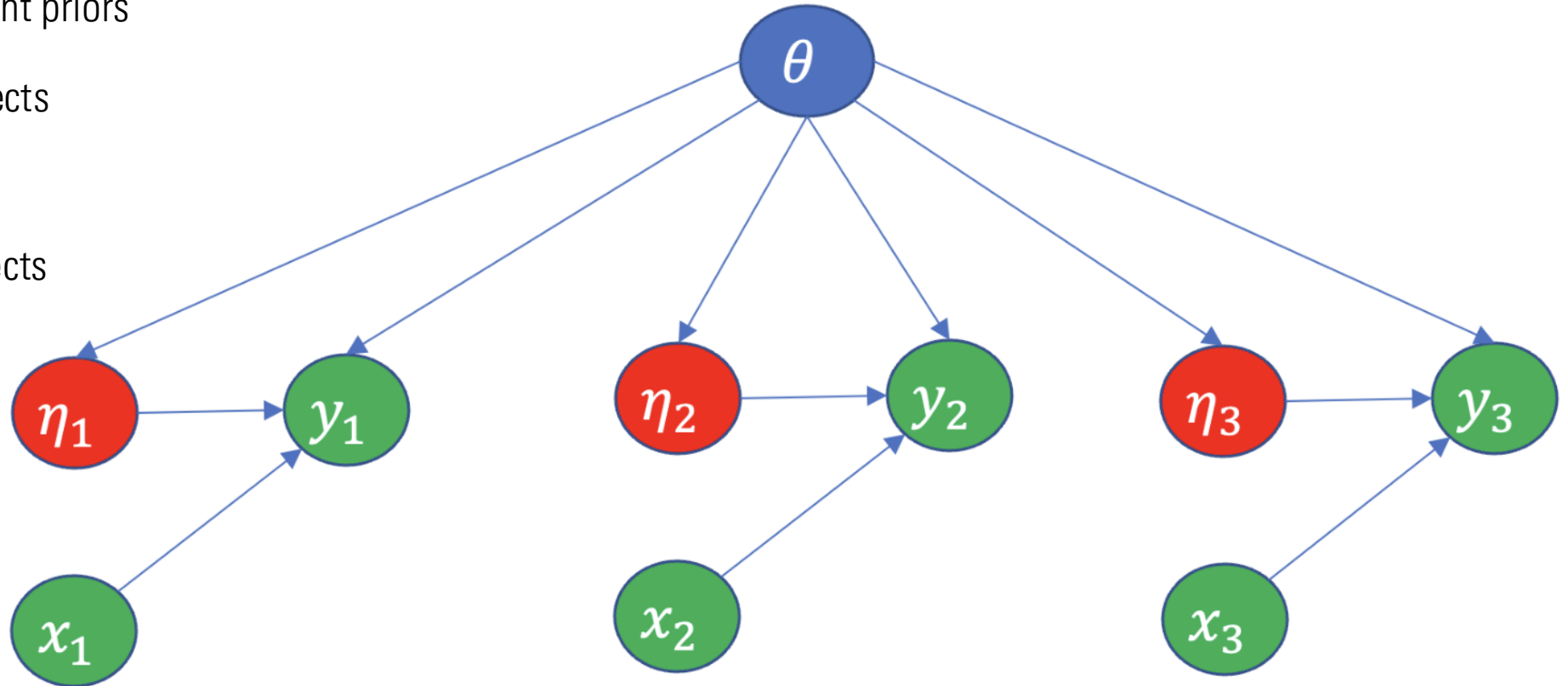
- Fixed effects, θ
 - Model parameters modelled as deterministic quantities
- Random effects, η
 - Model parameters modelled as random variables

Hierarchical

We typically define hierarchies where θ are shared parameters but η is subject-specific.



- θ – all the population parameters.
 - Shared across subjects
- η – random effects of all subjects.
 - η_1 specific to subject 1
 - Typically has θ -dependent priors
- x – covariates of all the subjects
 - x_1 specific to subject 1
- y – responses of all the subjects
 - y_1 specific to subject 1





In a Pumas model

```
@model begin
  @param begin
     $\theta$  {
       $\theta \in \text{VectorDomain}(4, \text{lower} = \text{zeros}(4))$ 
       $\Omega \in \text{PSDDomain}(2)$ 
       $\Sigma \in \text{RealDomain}(\text{lower} = 0.0)$ 
       $a \in \text{RealDomain}(\text{lower} = 0.0, \text{upper} = 1.0)$ 
    }
  end
   $\eta_i | \theta$  {
    @random begin
       $\eta \sim \text{MvNormal}(\Omega)$ 
    end
  }
   $x_i$  {
    @covariates sex wt etn
  }
   $y_i | \theta, \eta_i, x_i$  {
    @pre begin
       $\theta_1 := \theta[1]$ 
       $K_a = \theta_1$ 
       $CL = \theta[2] * ((wt / 70)^{0.75}) * (\theta[4]^{\text{sex}}) * \exp(\eta[1])$ 
       $V_c = \theta[3] * \exp(\eta[2])$ 
    end
    @dynamics begin
      Depot' = -Ka * Depot
      Central' = Ka * Depot - (CL / Vc) * Central
      Res' = Depot - Central
    end
    @derived begin
      conc = @. Central / Vc
      dv ~ @. Normal(conc, conc *  $\Sigma$ )
      T_max = maximum(t)
    end
    @observed begin
      obs_cmax = maximum(dv)
    end
  end
end
```



Don't assign too much meaning to the random effects

- Indicates unknown parameters that vary between subjects (or whatever hierarchy we use)
- Usually tied very closely to a fixed effect in pharmacometrics. $CL = tvCL \cdot \exp(\eta_{cl})$.
- Enables degrees of freedom along which the model can account for outcome heterogeneity.



Random effects during simulation?

Simple
Just sample and use

θ

$\eta_i | \theta$

x_i

$y_i | \theta, \eta_i, x_i$

```
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  @param begin
     $\theta \in \text{VectorDomain}(4, \text{lower} = \text{zeros}(4))$ 
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     $\Sigma \in \text{RealDomain}(\text{lower} = 0.0)$ 
     $a \in \text{RealDomain}(\text{lower} = 0.0, \text{upper} = 1.0)$ 
  end
  @random begin
     $\eta \sim \text{MvNormal}(\Omega)$ 
  end
  @covariates sex wt etn
  @pre begin
     $\theta_1 := \theta[1]$ 
     $Ka = \theta_1$ 
     $CL = \theta[2] * ((wt / 70)^{0.75}) * (\theta[4]^{\text{sex}}) * \exp(\eta[1])$ 
     $Vc = \theta[3] * \exp(\eta[2])$ 
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    Depot' = -Ka * Depot
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    dv ~ @. Normal(conc, conc *  $\Sigma$ )
    T_max = maximum(t)
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  @observed begin
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  end
end
```



Fitting with random effects

Their effect is largely determined by how they contribute to the loss function of a model fit

Joint likelihood (?)

The likelihood of both θ and η under the observations y and covariates x

$$L(\theta, \eta \mid y, x)$$

Related to $p(y \mid \theta, \eta, x)$ - the probability of observing y given a specific value of θ and η .

Fit model by simply finding the values of θ and η that maximizes this joint likelihood?



Fitting with random effects

Their effect is largely determined by how they contribute to the loss function of a model fit

Marginal Likelihood (!)

Integrates out the effect of the random effects

$$L(\boldsymbol{\theta} \mid \mathbf{y}, \mathbf{x}) = \int L(\boldsymbol{\theta}, \boldsymbol{\eta} \mid \mathbf{y}, \mathbf{x}) \cdot p_{\text{prior}}(\boldsymbol{\eta} \mid \boldsymbol{\theta}) d\boldsymbol{\eta}$$

Average likelihood of fixed effects over the random effect prior.

Conversely, average probability of $\mathbf{y}$ under a given $\boldsymbol{\theta}$ and $\mathbf{x}$ – averaged over the probable (under the prior) $\boldsymbol{\eta}$.

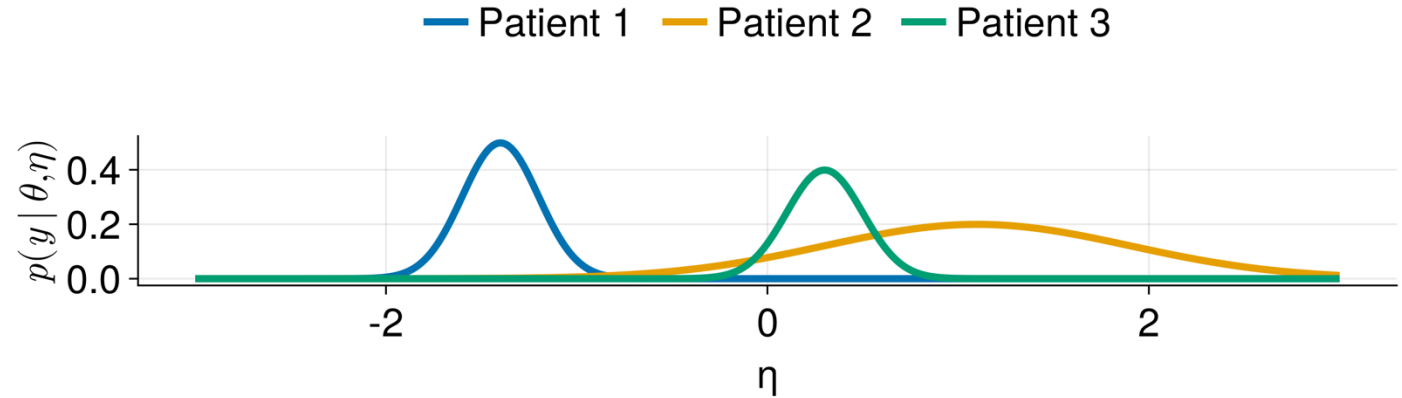
Different methods/approximations: Laplace, FOCE and EM



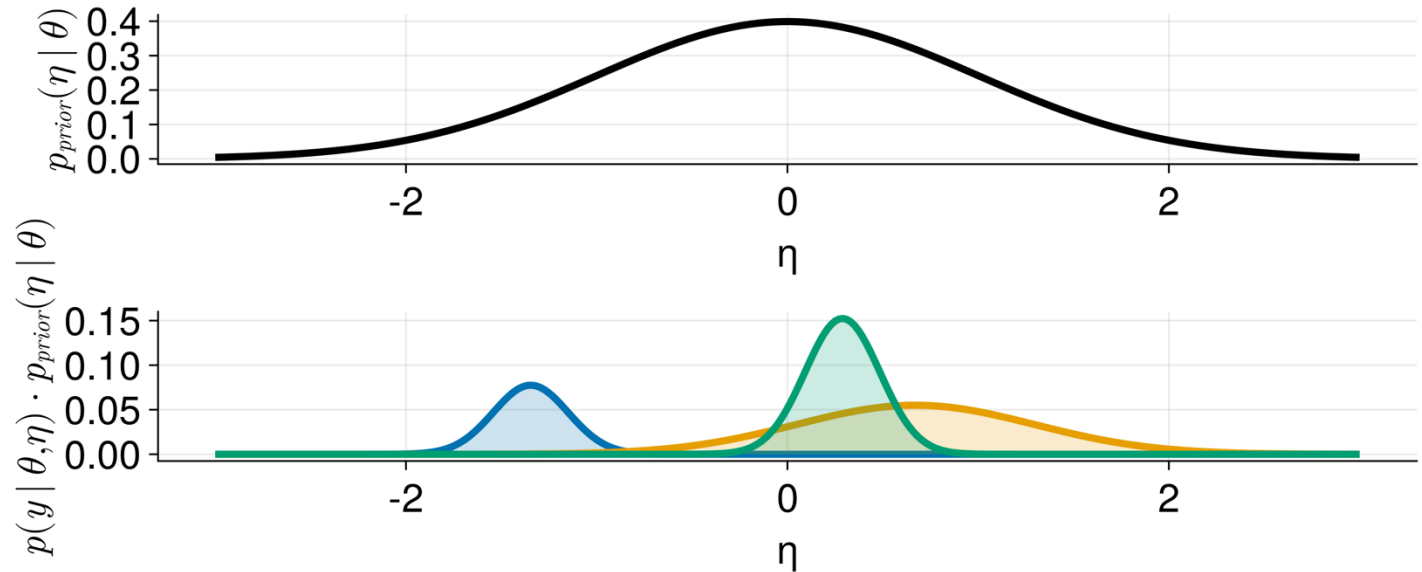
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Conditional likelihood



Marginal likelihood





Fitting algorithms in DeepPumas

	Marginal likelihood	Joint likelihood
Prior / regularization on θ	MAP(FO()) MAP(FOCE()) MAP(LaplaceI())	JointMAP()
No prior / regularization on θ	FO() FOCE() LaplaceI()	N/A



But what “are” the random effects?

- What do they capture?
- What happens when you use random effects as part of your NN input?
- Let's see in an exercise!